Temperature & Humidity Sensor RS485 Modbus RTU

ES35-SW

USER MANUAL

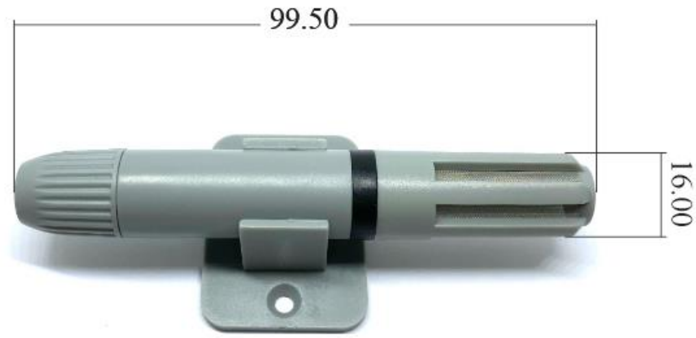
1. Description:

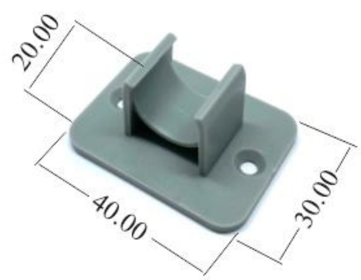
- ES35-SW is a precision temperature & humidity sensor using MODBUS RTU Protocol, easy to build cost-effective temperature and humidity degree real-time remote monitoring system.
- Use chip SHT35 with high accuracy and this probe is of reliable quality, stable performance.
- Internal sealant design and stainless probe makes it waterproof, damp resistant and not easy to rust.
- Use switching power, avoid heat generation when used at high voltage.
- Setup modbus address by hardware (switch) or software (PC), make installation easier.

2. Specitifications:

Power Supply		DC 8V – 30V
Use Range	Temperature	-20 ~ 80°C
	Humidity	0 ~ 100% RH
Accuracy	Temperature	±0.2°C
	Humidity	±1.5% RH
Performance		< 0.1W
Communication		RS485 MODBUS RTU
Dimension		99.5x16mm
Weight		~19 g

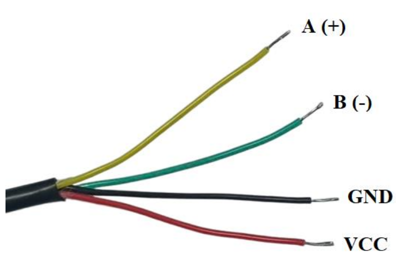
3. Dimensions:





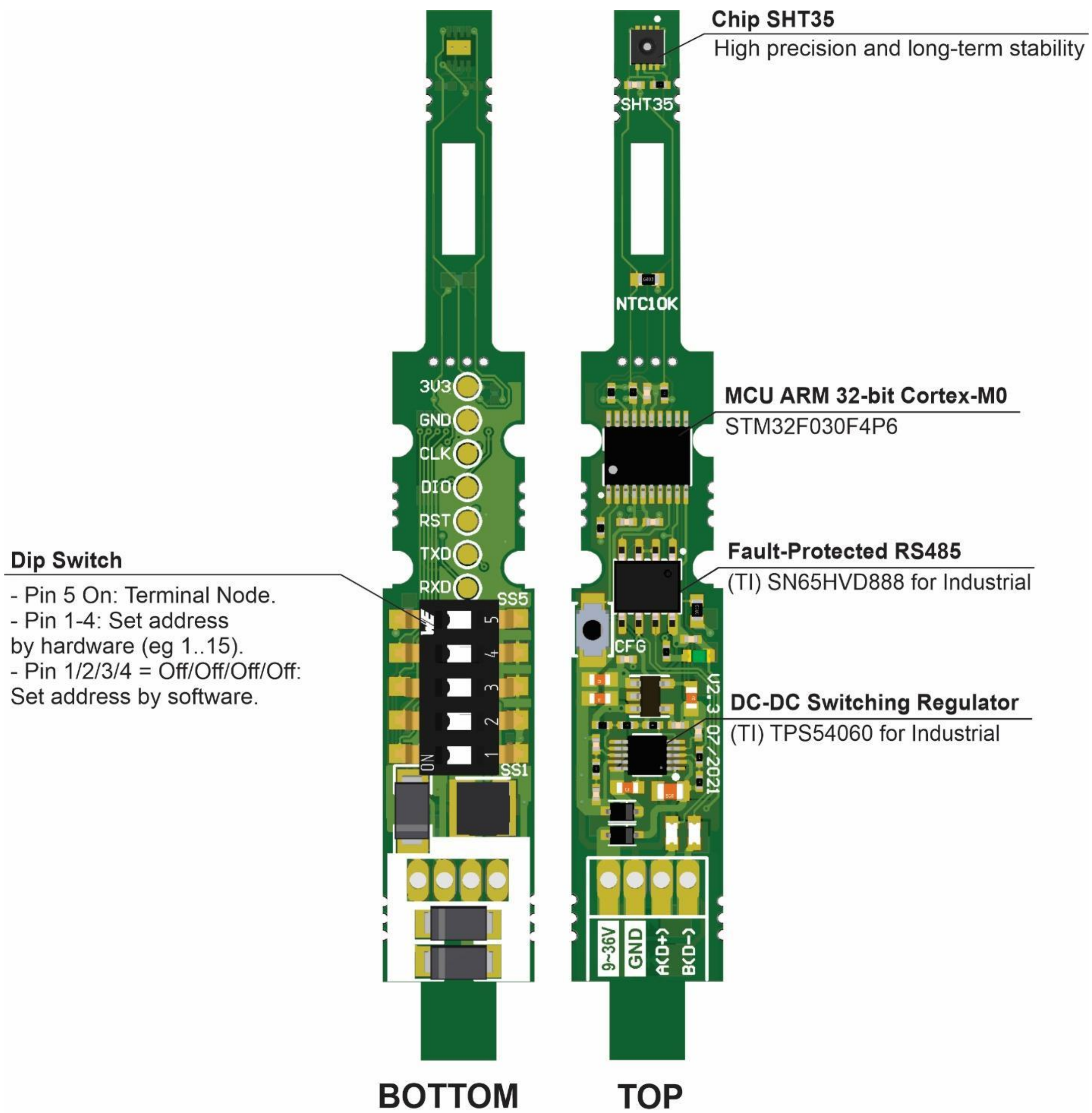
Unit (mm)

4. Output Leads:



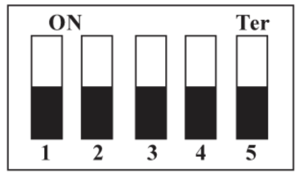
	Wire Color	Name	Connect
1	Black	GND	GND
2	Red	VCC	Power DC (8 – 30V)
3	Yellow	A +	RS485 Pin A +
4	Green	B -	RS485 Pin B -

5. Inside Sensor ES35-SW



USE CHIP FROM TEXAS INSTRUMENTS (TI)

6. Install Dip Switch (DP1)



DP1

Install Terminal Node	
DP1 Pin 5: On DP1 Pin 5: Off	Terminal Resistor 120 Ohm. Not Install Terminal Resistor
Install Address Modbus	
DP1 Pin 1/2/3/4	When modbus address is set (eg 1..15), then modbus RS485 communication is enabled. (Hardware Address)
Off/Off/Off/Off	Install modbus address by software, default address = 1.
On/Off/Off/Off Off/On/Off/Off … On/On/On/On	Address = 1 Address = 2 Address = 15

Attention:

After changed DP1 Pin 4/3/2/1, re-power device to accept the address.

When DP1 Pin 4/3/2/1 Off/Off/Off/Off, change to mode install Address by App on the PC(Software), Default Address = 1.

7. Communication Protocol – RS485 MODBUS RTU

Parameter Setting Modbus

❖ **Default Communication Parameter:**

Address (ID)	Baudrate	Data Bits	Parity	Stop Bit
1	9600	8	None	1

❖ **Register Table:**

Address	Name	Type	Function	Description
0	Temperature	Read Only	3	Value: -20 - 80(°C)
1	Humidity	Read Only	3	Value: 0 - 100(%)
100	Address Device(ID)	Read/Write	3, 6	+ Install by switch DP1: 1-15. + Install by software: 1-247.
101	Baudrate	Read/Write	3, 6	2: 4800 3: 9600 4: 14400 5: 19200 6: 38400 7: 56000 8: 57600 9: 115200
102	Parity	Read Only	3	0: None Parity
103	DataBits	Read Only	3	8: 8 Databits
104	StopBits	Read Only	3	1: 1 Stop bit
105	Setting Mode	Read Only	3	1: Mode setup address by Switch DP1 (Hardware) 0: Mode setup address by Software
106	Temperature Correction	Read/Write	3, 6	Giá trị: -10 ~ 10 (°C)
107	Humidity Correction	Read/Write	3, 6	Giá trị: -10 ~ 10 (%)

❖ **Communication Commands:**

Functions is used:

- 3: Read Holding Registers
- 6: Write Single Register

➢ **(Function 3) Read Holding Registers**

Read value of registers. Address and value of resgister are described on register table.

Example 1: Read 2 registers Temperature & Humidity from device modbus with address 1.

Address register of temperature is 0 (0x00).

Address register of humidity is 1 (0x01).

1. *Send command format:*

Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x03
3	Start Address(Upper)	0x00
4	Start Address(Lower)	0x00
5	Number of Registers(Upper)	0x00
6	Number of Registers(Lower)	0x02
7	Check CRC(Lower)	0xC4
8	Check CRC(Upper)	0x0B

2. *Receive command format:*

Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x03
3	Data Size(Bytes)	0x04
4	Value Temperature(Upper)	0x01
5	Value Temperature(Lower)	0x25
6	Value Humidity(Upper)	0x01
7	Value Humidity (Lower)	0xAE
8	Check CRC(Lower)	0x6A
9	Check CRC(Upper)	0x28

3. *Calculating Data:*
    - Divide value temperature & humidity by 10
      => Value Temperature (°C), Value Humidity(% RH).
    - Value temperature respond: 0x0125

    The value 0x0125 is converted to the decimal = 293.
    After using 293 ÷ 10 = 29,3 => Value temperature 29,3(°C).
    - Value humidity respond: 0x01AE

    The value 0x01AE is converted to the decimal = 430.
    After using 430 ÷ 10 = 43,0 => Value humidity 43,0(% RH).

Example 2: Read 6 registers of setting parameters starting from address register 100(0x64) with device address = 1.

(Device address, Baudrate, Parity, Databits, Stopbits, Setting Mode)

1. *Send command format:*

Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x03
3	Start Address(Upper)	0x00
4	Start Address(Lower)	0x64
5	Number of Registers(Upper)	0x00
6	Number of Registers(Lower)	0x06
7	Check CRC(Lower)	0x84
8	Check CRC(Upper)	0x17

2. *Receive command format:*

Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x03
3	Data Size(Bytes)	0x0C
4	Value Address(Upper)	0x00
5	Value Address(Lower)	0x01
6	Value Baudrate(Upper)	0x00
7	Value Baudrate(Lower)	0x03
8	Value Parity(Upper)	0x00
9	Value Parity(Lower)	0x00
10	Value Databits(Upper)	0x00
11	Value Databits(Lower)	0x08

12	Value Stopbits(Upper)	0x00
13	Value Stopbits(Lower)	0x01
14	Value Setting Mode(Upper)	0x00
15	Value Setting Mode(Lower)	0x00
16	Check CRC(Lower)	0x33
17	Check CRC(Upper)	0x7D

3. *Detail Data:*

Value Address (Register 100): 0x0001 => Slave Address = 1

Value Baudrate (Register 101): 0x0003 => 3: 9600

Value Parity (Register 102): 0x0000 => 0: None Parity

Value DataBits (Register 103): 0x0008 => 8: 8 Databits

Value StopBits (Register 104): 0x0001 => 1: 1 Stop bit

Value Setting Mode (Register 105): 0x0000 => 0: Install Slave Address by Software.

- **(Function 6) Write Single Register**
  Write value setting to registers. Address and value of register are described on register table.
  Attention: Command write address device is only applied in software address setting mode (Setting Mode = 0).

Example 1: Write new address device by write in register 100(0x64).
Current Address: 1.
Address want to change: 10 (New address)

1. *Send command format:*

Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x06
3	Start Address(Upper)	0x00
4	Start Address(Lower)	0x64
5	Data address Write(Upper)	0x00
6	Data address Write(Lower)	0x0A
7	Check CRC(Lower)	0x48
8	Check CRC(Upper)	0x12

2. *Receive command format:*

Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x06
3	Start Address(Upper)	0x00
4	Start Address(Lower)	0x64
5	Data address Write(Upper)	0x00
6	Data address Write(Lower)	0x0A
7	Check CRC(Lower)	0x48
8	Check CRC(Upper)	0x12

3. *Detail Data:*

Start Address: 0x0064 => 100

Data Address Write: 0x000A => 10

Example 2: Change Baudrate device by write in register 101 (0x65).

Current Baudrate 3: 9600 => Baudrate want to change 5: 19200 (New Baudrate)

1. *Send command format:*

Byte	Mô tả chức năng	Example
1	Slave Address	0x01
2	Function Code	0x06
3	Start Address(Upper)	0x00
4	Start Address(Lower)	0x65
5	Data Baudrate Write(Upper)	0x00
6	Data Baudrate Write(Lower)	0x05
7	Check CRC(Lower)	0x59
8	Check CRC(Upper)	0xD6

2. *Receive command format:*

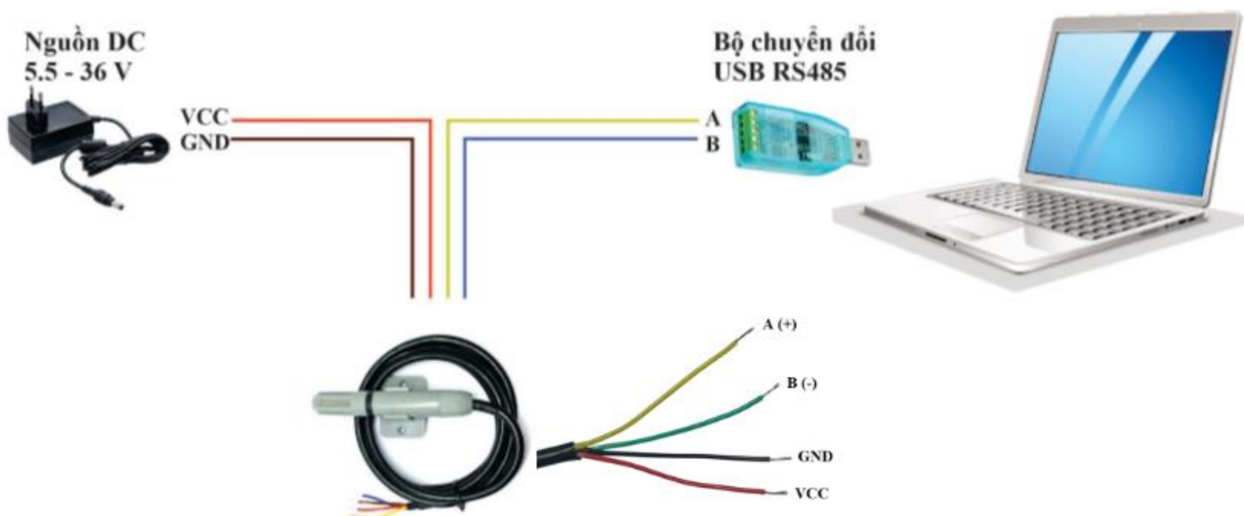
Byte	Funtion Description	Example
1	Slave Address	0x01
2	Function Code	0x06
3	Start Address(Upper)	0x00
4	Start Address(Lower)	0x65
5	Data Baudrate Write(Upper)	0x00
6	Data Baudrate Write(Lower)	0x05
7	Check CRC(Lower)	0x59
8	Check CRC(Upper)	0xD6

3. *Detail Data:*

Start Address:	0x0065	=> 101
Data Baudrate Write:	0x0005	=> 5: 19200

Attention: *Settings will be changed after re-power device.*

8. Connect device to PC:



Connect device with computer

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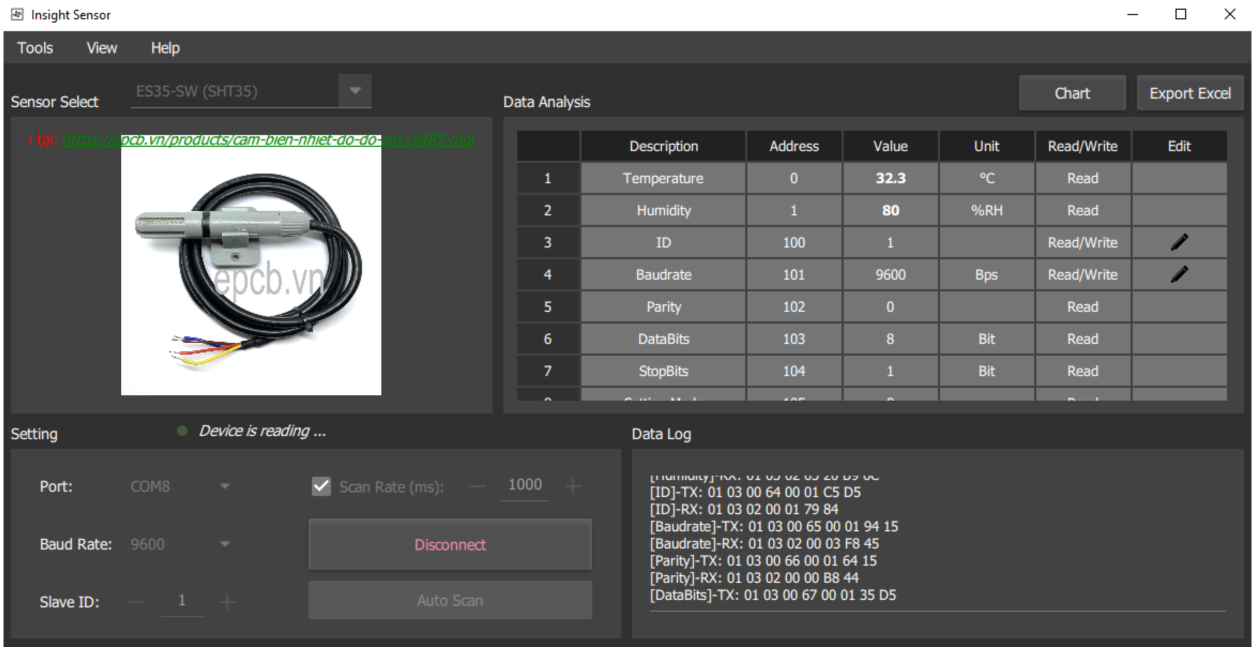
❖ **Connect device with APP PC** (*Insight Sensor)*

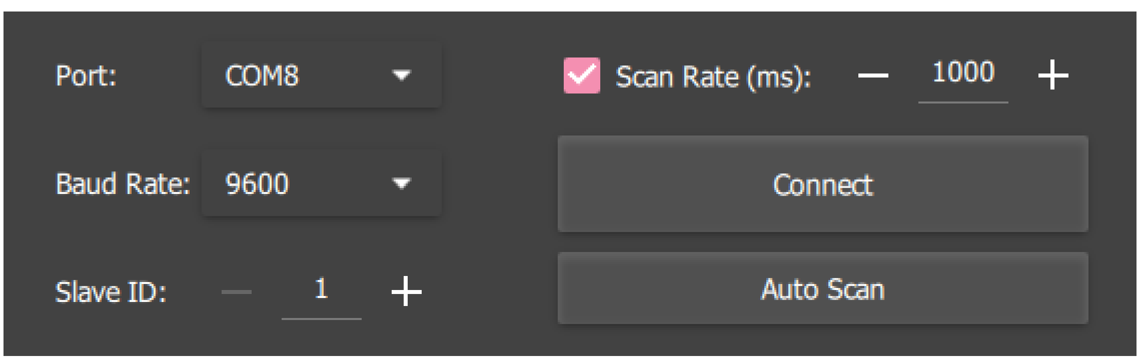
- Using App "*Insight Sensor*" (attach file).



App on PC

Step 1: Connect device to PC

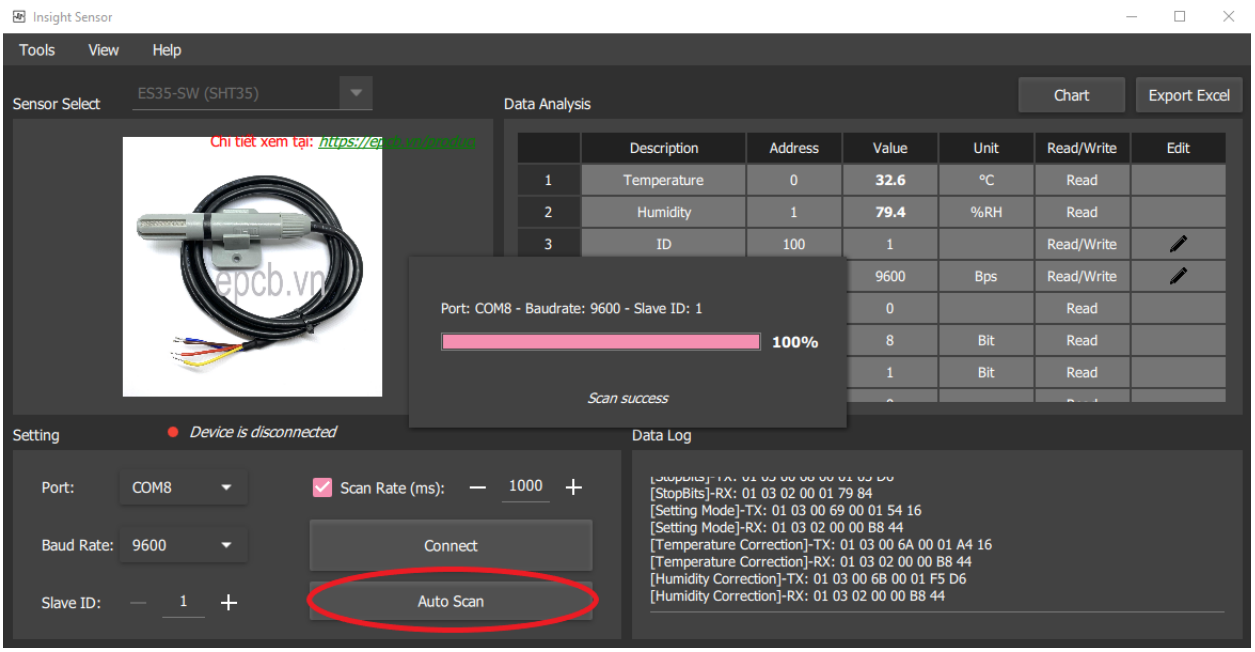
Step 2: Choose COM, setting parameters and connect.



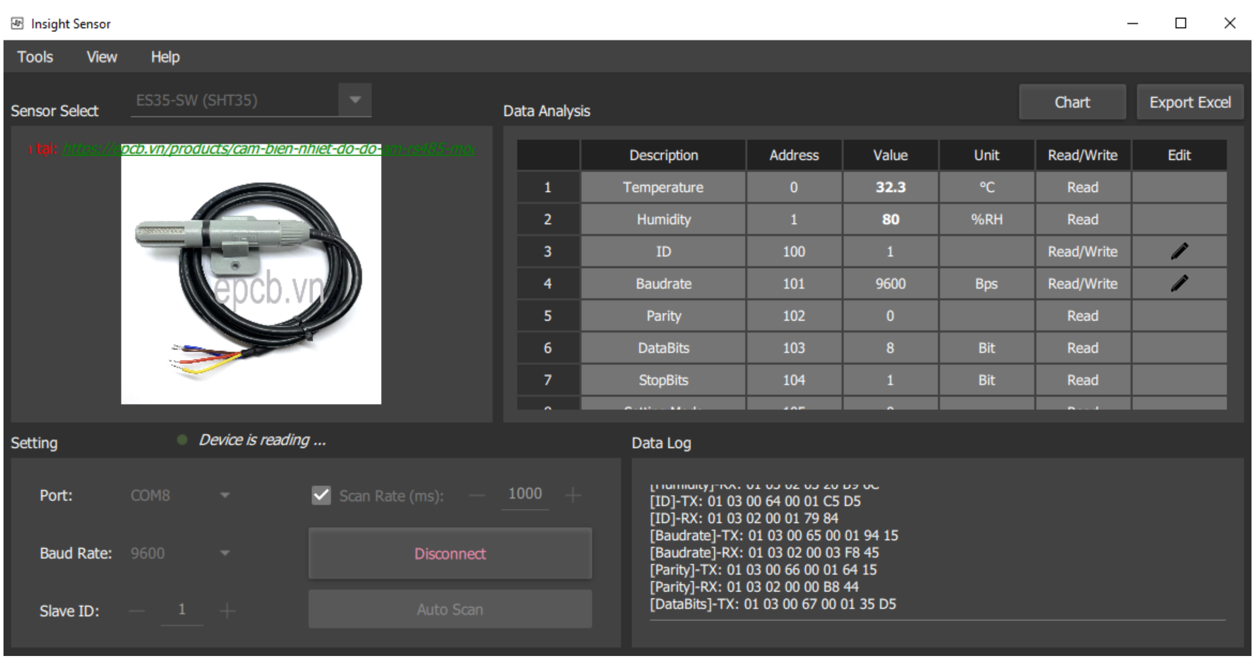
Quick Setup

Auto Scan:

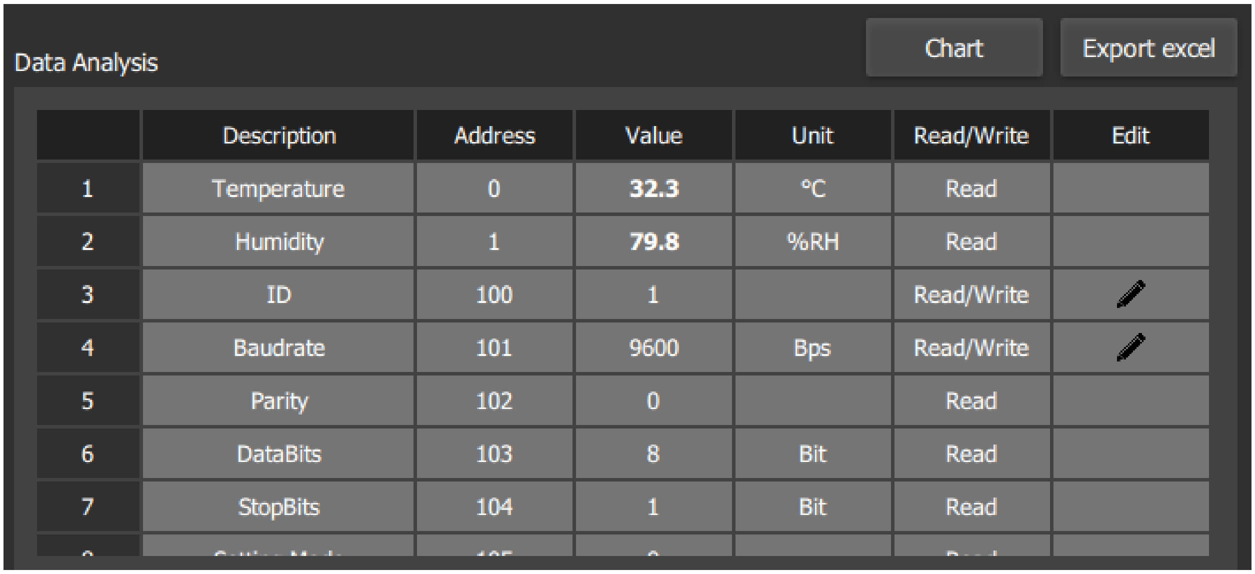
Choose COM -> Push Button Auto Scan -> App auto scan parameters and connect to sensor device.



Connect success:



Mode View Data Table and Chart

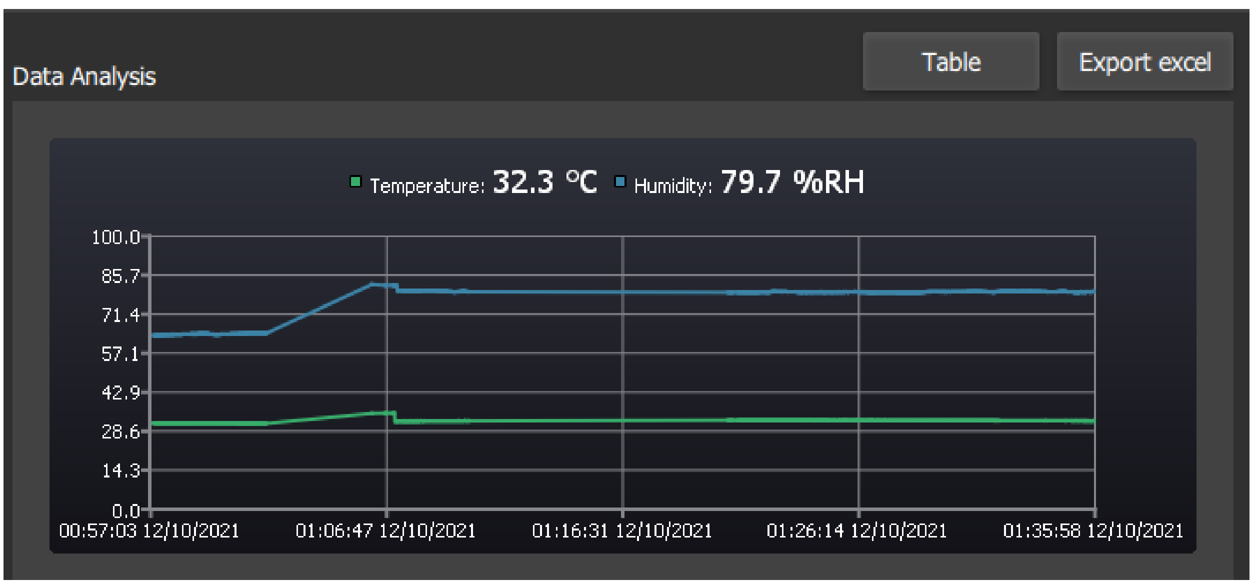


Data Analysis

Chart | Export excel

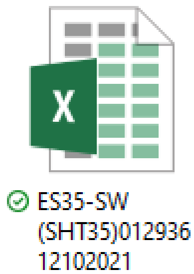
	Description	Address	Value	Unit	Read/Write	Edit
1	Temperature	0	32.3	°C	Read	
2	Humidity	1	79.8	%RH	Read	
3	ID	100	1		Read/Write	
4	Baudrate	101	9600	Bps	Read/Write	
5	Parity	102	0		Read	
6	DataBits	103	8	Bit	Read	
7	StopBits	104	1	Bit	Read	

Data View Table



Data View Chart

Export Excel: Save Temperature and Humidity Value to File Excel



	A	B	C	D	E
1	12/10/2021	00:57:09	63.7	%RH	
2	12/10/2021	00:57:11	63.7	%RH	
3	12/10/2021	00:57:13	63.6	%RH	
4	12/10/2021	00:57:14	63.7	%RH	
5	12/10/2021	00:57:16	63.7	%RH	
6	12/10/2021	00:57:18	63.7	%RH	
7	12/10/2021	00:57:19	63.7	%RH	
8	12/10/2021	00:57:21	63.7	%RH	
9	12/10/2021	00:57:23	63.7	%RH	
10	12/10/2021	00:57:25	63.7	%RH	
11	12/10/2021	00:57:26	63.7	%RH	
12	12/10/2021	00:57:28	63.7	%RH	
13	12/10/2021	00:57:30	63.7	%RH	
14	12/10/2021	00:57:31	63.8	%RH	
15	12/10/2021	00:57:33	63.8	%RH	
16	12/10/2021	00:57:35	63.8	%RH	
17	12/10/2021	00:57:36	63.8	%RH	
18	12/10/2021	00:57:38	63.8	%RH	
19	12/10/2021	00:57:40	63.8	%RH	
20	12/10/2021	00:57:41	63.9	%RH	
21	12/10/2021	00:57:43	63.9	%RH	
22	12/10/2021	00:57:45	63.9	%RH	
23	12/10/2021	00:57:46	63.9	%RH	

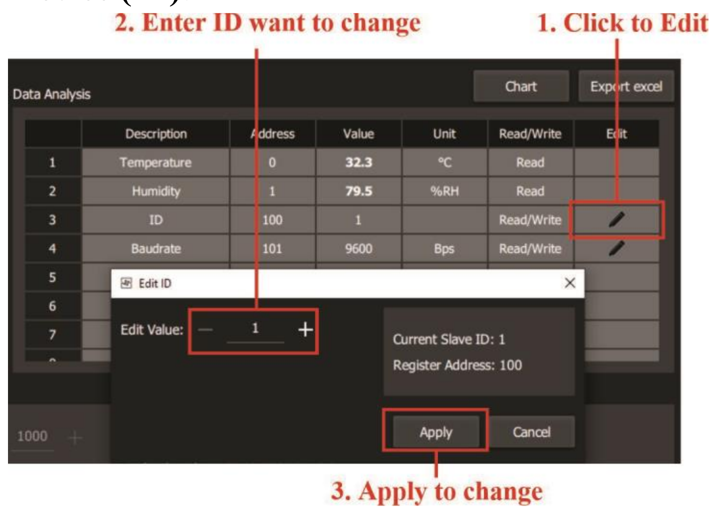
Temperature | **Humidity**

	A	B	C	D	E
1	12/10/2021	00:57:09	31.5	°C	
2	12/10/2021	00:57:11	31.5	°C	
3	12/10/2021	00:57:13	31.5	°C	
4	12/10/2021	00:57:14	31.5	°C	
5	12/10/2021	00:57:16	31.5	°C	
6	12/10/2021	00:57:18	31.5	°C	
7	12/10/2021	00:57:19	31.5	°C	
8	12/10/2021	00:57:21	31.5	°C	
9	12/10/2021	00:57:23	31.5	°C	
10	12/10/2021	00:57:24	31.5	°C	
11	12/10/2021	00:57:26	31.5	°C	
12	12/10/2021	00:57:28	31.5	°C	
13	12/10/2021	00:57:29	31.5	°C	
14	12/10/2021	00:57:31	31.5	°C	
15	12/10/2021	00:57:33	31.5	°C	
16	12/10/2021	00:57:35	31.4	°C	
17	12/10/2021	00:57:36	31.5	°C	
18	12/10/2021	00:57:38	31.5	°C	
19	12/10/2021	00:57:40	31.5	°C	
20	12/10/2021	00:57:41	31.5	°C	
21	12/10/2021	00:57:43	31.5	°C	
22	12/10/2021	00:57:45	31.5	°C	
23	12/10/2021	00:57:46	31.5	°C	

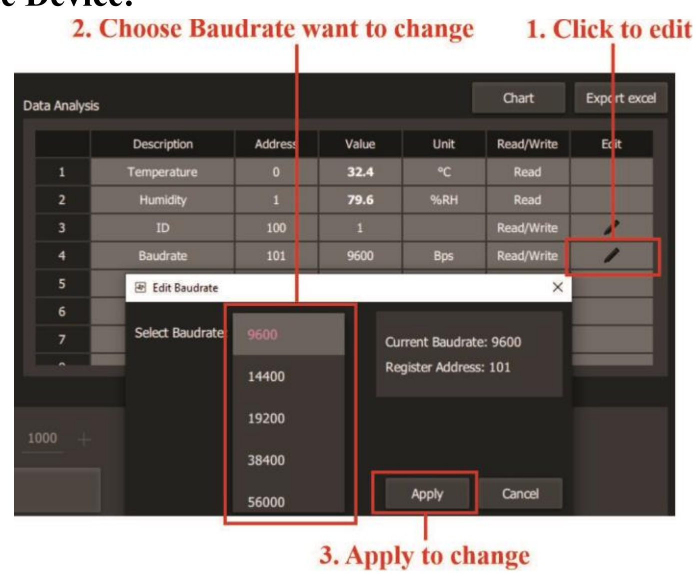
Temperature | Humidity

Step 3: Write registers, setting new parameters for sensor device.

- **Setting Address Device (ID):**



- **Setting Baudrate Device:**



Similar for remain parameters.

Settting Mode: 0 (Software) mode install address device by Software.

Settting Mode: 1 (Hardware) mode install address device by Switch in Hardware. Can't write address by Software.

(DP1 Pin 4/3/2/1 Off/Off/Off/Off)

Attention**:** *Settings will be changed after re-power device.*

You can use Apps Modbus on PC as Modbus Poll,..

To read & write parameters device base on register table.

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